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Project Title: Digital Attendance System

Submitted By

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ABSTRACT

Attendance management plays a vital role in maintaining academic discipline and ensuring student participation in educational institutions. Manual attendance systems are timeconsuming, prone to human error, and difficult to maintain over long periods. To address these challenges and contribute toward SDG 4 – Quality Education, a Smart Attendance System was designed and implemented using Arduino technology.

This project presents a microcontroller-based digital attendance system developed using an Arduino UNO, 16×2 LCD display, and 4×4 matrix keypad. The system allows the user to initialize attendance by entering the total number of students using the keypad. Each student's roll number is entered individually, and attendance is marked as Present or Absent using predefined keypad keys. The system stores and processes attendance data in real-time.

After all student records are entered, the system automatically calculates the overall attendance percentage using programmed logic and displays the final result on the LCD screen. The design ensures simplicity, low cost, ease of implementation, and efficient data handling. The program utilizes structured conditional statements, counters, and percentage calculation algorithms to ensure accurate results.

The proposed Smart Attendance System reduces manual workload, minimizes errors, and provides instant attendance analysis. It is especially useful in classrooms, training centers, and small educational institutions where automation can significantly improve efficiency. This project demonstrates how embedded systems can be applied effectively to enhance academic administration and support the global objective of ensuring inclusive and quality education for all.

TABLE CONTENTS

SL.NO	CONTENTS	PAGE NO
1	INTRODUCTION	4
2	PROBLEM STATEMENT	5
3	OBJECTIVES	6
4	METHODOLOGY	7
5	COMPONENTS USED	8-19
6	RESULTS	20-21
7	SDG IMPACT ANALYSIS	22
8	SUMMARY	23
9	DISCUSSION	24-25

CHAPHER 1

INTRODUCTION

Attendance monitoring is an essential part of academic institutions, as it reflects student participation, discipline, and engagement in learning activities. Traditionally, attendance is recorded manually by teachers using paper registers. Although simple, this method consumes valuable teaching time, increases the possibility of human errors, and makes long-term data management difficult. In large classrooms especially, manual attendance recording can significantly reduce instructional time and administrative efficiency.

With advancements in embedded systems and microcontroller technologies, automation has become a practical solution for improving educational processes. Integrating electronics into administrative tasks enhances accuracy, speed, and reliability. In this context, the Smart Attendance System was developed as a low-cost, efficient, and easy-to-implement solution using Arduino technology.

The proposed system is designed using an Arduino UNO microcontroller, a 16×2 LCD display for user interaction, and a 4×4 matrix keypad for data entry. The system allows the user to begin attendance by initializing the total number of students. Each student's roll number is entered using the keypad, and attendance is marked as Present or Absent through designated keys. The system maintains a count of present students and total entries, and once all students are recorded, it calculates and displays the final attendance percentage on the LCD screen.

This project contributes to SDG 4 – Quality Education, which aims to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. By reducing manual workload and minimizing errors, the Smart Attendance System supports better classroom management and effective academic monitoring.

The system demonstrates how embedded systems can be applied in real-world educational environments to automate routine tasks. It is particularly suitable for schools, colleges, and training centers that require a simple and economical attendance management solution. The project also highlights the importance of programming logic, user interface design, and hardware-software integration in solving practical problems.

Overall, the Smart Attendance System represents a step toward digital transformation in education, making attendance tracking more accurate, efficient, and reliable.

CHAPTER 2

PROBLEM STATEMENT

Attendance management is a critical administrative function in educational institutions. Conventional attendance recording methods rely on manual entry in physical registers, which is time-consuming, error-prone, and inefficient, particularly in classrooms with a large number of students. Manual calculation of attendance percentages further increases the possibility of computational errors and requires additional effort from faculty members. Moreover, maintaining paper-based records limits data accessibility and long-term storage efficiency.

The absence of automation in traditional systems results in reduced accuracy, delayed data processing, and inefficient classroom management. Therefore, there is a need for a reliable, low-cost, and automated attendance solution that minimizes human intervention, ensures accurate data recording, and provides real-time attendance percentage calculation. The proposed Arduino-based Smart Attendance System addresses these limitations by integrating a keypad for roll number entry, an LCD for real-time display, and programmed logic for automated computation, thereby supporting efficient academic administration in alignment with SDG 4 – Quality Education.

CHAPTER 3

OBJECTIVES

- To design and develop a microcontroller-based digital attendance system using Arduino UNO, keypad, and LCD display.
- To automate attendance recording by allowing roll number entry and marking students as Present or Absent digitally.
- To reduce manual errors associated with traditional paper-based attendance systems.
- To calculate attendance percentage automatically using programmed logic and display the result instantly.
- To minimize time consumption in classroom attendance management.
- To develop a low-cost and easy-to-implement solution suitable for schools, colleges, and training centers.
- To support SDG 4 – Quality Education by improving efficiency and accuracy in academic administration.

CHAPTER 4

METHODOLOGY

The methodology of the Digital Attendance System involves the design, development, and implementation of a microcontroller-based automated attendance recording system using Arduino technology. The overall process is divided into hardware setup, software development, and system operation.

Initially, the required hardware components such as Arduino UNO, 16×2 LCD display (I2C module), and 4×4 matrix keypad were selected. The keypad was connected to the digital input pins of the Arduino to enable roll number entry and attendance marking. The LCD display was interfaced using the I2C communication protocol to reduce wiring complexity and display real-time messages to the user.

After completing the hardware connections, the software program was developed using the Arduino IDE. Necessary libraries such as Wire.h, LiquidCrystal_I2C.h, and Keypad.h were included to enable communication between components. The program logic was structured into three main stages:

1. **Initialization Stage** – The system waits for the user to start the process and enter the total number of students.
2. **Attendance Recording Stage** – Roll numbers are entered through the keypad, and attendance is marked as Present or Absent using assigned keys. The system keeps track of total entries and present count.
3. **Result Calculation Stage** – After recording all students, the system calculates the attendance percentage using the formula:

Attendance%=No of present students/Total studentsx100

The final percentage is displayed on the LCD screen.

Conditional statements and counters were used to control system flow and ensure accurate data processing. The system was tested multiple times to verify correct roll number input, attendance marking, and percentage calculation.

CHAPTER 5

COMPONENTS USED

1. Arduino Uno

The Arduino Uno is a series of open-source microcontroller board based on a diverse range of microcontrollers (MCU). It was initially developed and released by the Arduino company in 2010.The microcontroller board is equipped with sets of digital and analog input/output (I/O)pins that may be interfaced to various expansion boards (shields) and other circuits.[1] The board has 14 digital I/O pins (six capable of PWM output), 6 analog I/O pins, and is programmable with the Arduino IDE (Integrated Development Environment), via a type B USB cable.[4] It can be powered by a USB cable or a barrel connector that accepts voltages between 7 and 20 volts, such as a rectangular 9-volt battery. It has the same microcontroller as the Arduino Nano board, and the same headers as the Leonardo board. The hardware reference design is distributed under a Creative Commons Attribution Share-Alike 2.5 license and is available on the Arduino website. Layout and production files for some versions of the hardware are also available.

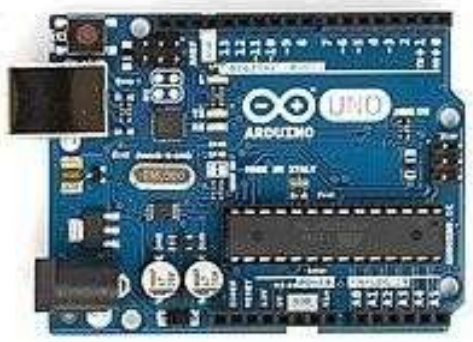

Fig 5.1

Technical specifications

- IC: Microchip ATmega328P (8-bit AVR core)
- Clock Speed: 16 MHz (on Uno R1 to R3 boards), though IC supports 20 MHz maximum at 5 Volts
- Flash memory: 32 KB, of which 0.5 KB used by the bootloader
- SRAM: 2 KB
- EEPROM: 1 KB
- USART peripherals: 1 (Arduino software default configures USART as a 8N1 UART)
- SPI peripherals: 1
- I^2C peripherals: 1
- Operating Voltage: 5 Volts (on Uno R1 to R3 boards), though IC supports 1.8 to 5.5 Volts (some 3rd-party Uno variations support 3.3V)

2. LCD

RG 1602 ,16x2 LCD Display in a captivating blue hue, perfect for your electronics projects. This versatile and reliable LCD screen features a 16x2 character layout, allowing for clear and crisp text display. Its compact design and easy-to-use interface make it ideal for a wide range of applications. Whether you're a hobbyist or a professional, this high-quality LCD display with a soothing blue backlight will enhance your projects and captivate your audience. Explore the possibilities today

This 16x2 Blue Jumbo Display, featuring the same captivating blue color but with a larger font size for enhanced visibility. This display offers 2 lines, each with 16 characters, capable of displaying both alphanumeric characters and numbers. Its versatile functionality makes it perfect for applications such as data scanners and meters, providing clear and easy-to-read indications of critical information.

Fig 5.2

Technical specifications

- Operating Voltage: 4.7V to 5.3V
- Operating Current 1mA (without backlight)
- Can display (16x2) 32 Alphanumeric Characters
- Custom Characters Support
- Works in both 8-bit and 4-bit Mode

3. Keypad

This 16-button keypad provides a useful human interface component for microcontroller projects. Convenient adhesive backing provides a simple way to mount the keypad in a variety of applications. The Keypad 4x4 features a total of 16 buttons in Matrix form. This is a membrane keypad with no moving parts. It has a nice overlay depicating a telephone type keypad with additional four functional buttons. A female 8-pin berg connector is provided for interfacing it with your mirocontroller circuits.

Fig 5.3

Technical specifications

- Maximum Rating: 24 VDC, 30 mA
- Interface: 8-pin access to 4x4 matrix
- Operating temperature: 32 to 122 °F (0 to 50°C)
- Dimensions: Keypad, 2.7 x 3.0 in (6.9 x 7.6 cm)
- Connector: Dupont 8 pins, 0.1" (2.54mm) Pitch
- Max. Circuit Rating: 24VDC, 30mA
- Insulation Spec.: 100M Ohm, 100V

5.2 CIRCUIT DIAGRAM

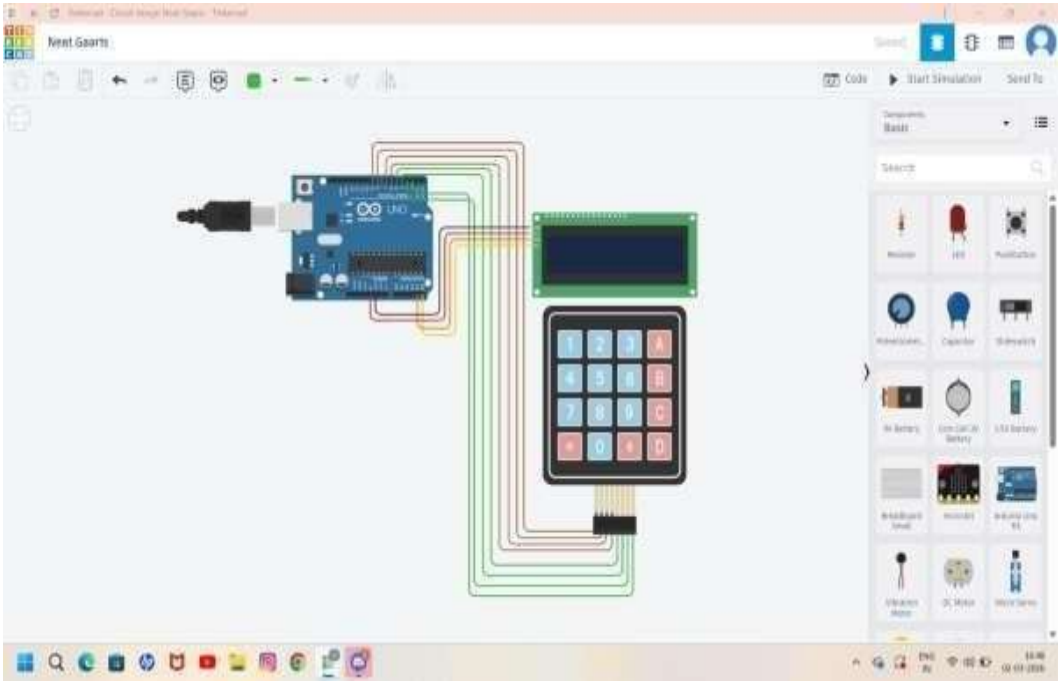

Fig 5.2.1

Circuit setup of a Digital Attendance System using Arduino UNO interfaced with a 16×2 LCD (I2C) and a 4×4 matrix keypad for digital attendance recording and percentage calculation.

The circuit diagram of the Digital Attendance System illustrates the interconnection between the Arduino UNO, 16×2 LCD display (I2C module), and 4×4 matrix keypad. The Arduino UNO acts as the central processing unit that controls data input, processing, and output display.

1. Arduino UNO (Microcontroller Unit)

The Arduino UNO is the core component of the system. It performs the following functions:

- Reads input signals from the keypad
- Executes programmed attendance logic
- Maintains counters for total students and present students
- Calculates attendance percentage
- Sends display data to the LCD

The board is powered through a USB connection or an external 5V power supply.

2. LCD Display (16×2 with I2C Module)

The 16×2 LCD is used to display system messages such as:

- "Press * to Start"
- "Enter No. of Students"
- Roll number display
- Present/Absent status
- Final attendance percentage

LCD Pin Connections (I2C Mode)

LCD Pin Connected To

VCC 5V (Arduino)

GND GND (Arduino)

SDA A4 (Arduino)

SCL A5 (Arduino)

Using the I2C module reduces the number of required wires and simplifies the circuit.

3. 4×4 Matrix Keypad

The keypad acts as the input device. It allows the user to:

- Enter total number of students
- Input roll numbers
- Mark attendance (A = Absent, B = Present)
- Start the system (*)
- Confirm entry (#)
- Display final result (D) Keypad Pin Connections The keypad consists of 4 row pins and 4 column pins connected to digital pins of Arduino:

Keypad Pin	Arduino Digital Pin
Row 1	9
Row 2	8
Row 3	7
Row 4	6
Column 1	5
Column 2	4
Column 3	3
Column 4	2

The Arduino scans rows and columns to detect key presses.

4. Power Connections

- Arduino is powered via USB or 5V adapter.
- LCD receives 5V and GND from Arduino.
- Keypad operates using digital signal connections and does not require a separate power supply.

5. Overall Circuit Operation

When the system is powered ON, the Arduino initializes the LCD and keypad. The keypad sends input signals to Arduino whenever a key is pressed. The Arduino processes the input, updates attendance counters, and displays relevant information on the LCD. After all students are recorded, the system calculates and displays the final attendance percentage.

The circuit design is simple, cost-effective, and suitable for educational institutions aiming to automate attendance management in support of SDG 4 – Quality Education.

5.3 FLOWCHART

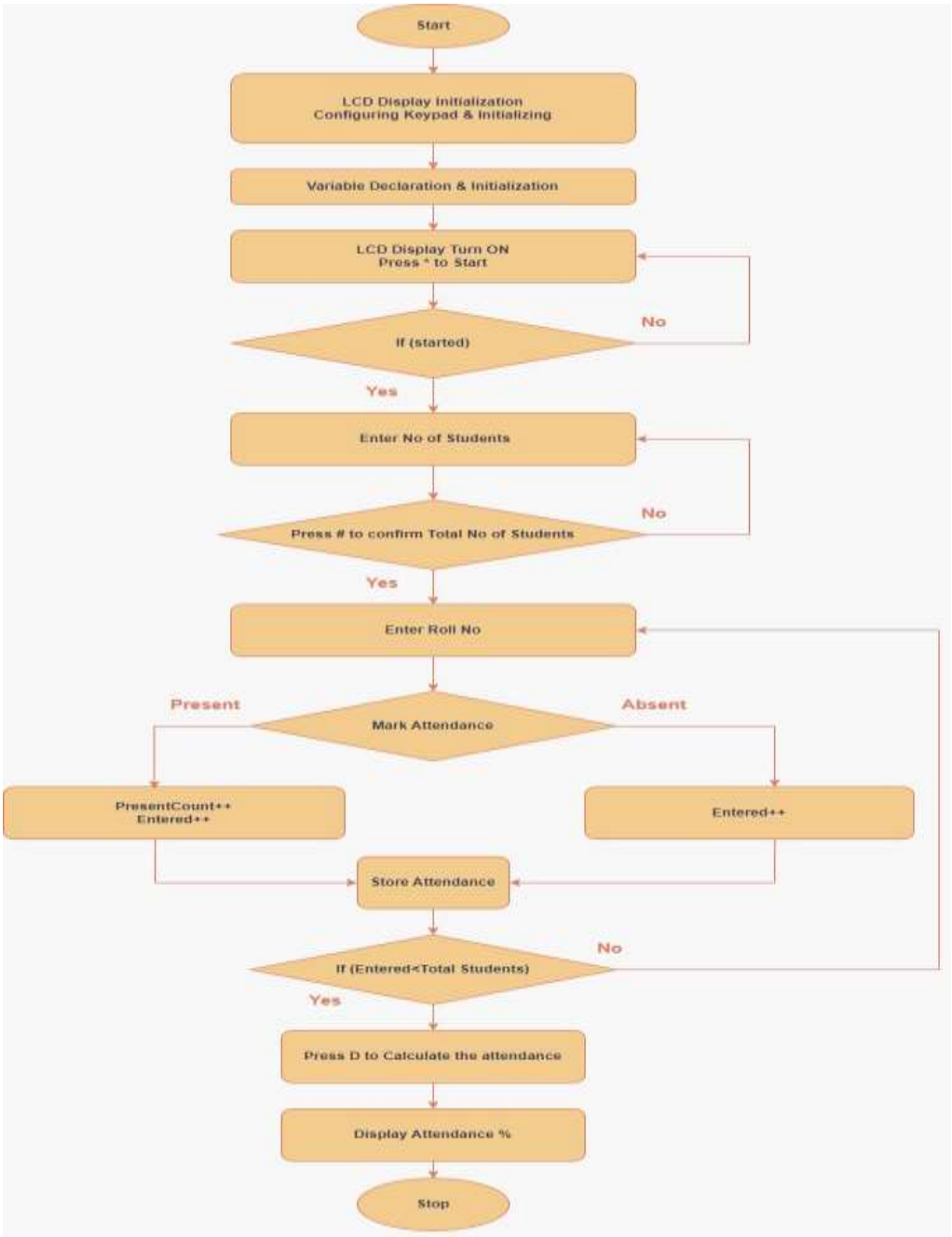

Fig.5.3.1

The flowchart represents the working process of the Digital Attendance System. The system starts by initializing the LCD display, configuring the keypad, and declaring necessary variables. It then prompts the user to press * to begin. After starting, the user enters the total number of students and confirms it by pressing #. The system then allows roll number entry for each student, where attendance is marked as Present or Absent. If marked present, the present counter and entered counter are

incremented; if absent, only the entered counter increases. The attendance data is stored, and the process repeats until the number of entered students equals the total number of students. Finally, the user presses D to calculate and display the overall attendance percentage on the LCD, and the system stops.

5.4 CODE

```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <Keypad.h>

LiquidCrystal_I2C lcd(0x27,16,2);   // change to 0x3F if needed

// -------- Keypad

Setup ---------const

byte ROWS = 4;

const byte COLS = 4;

char

keys[ROWS][COLS]

= {

  {'1','2','3','A'},

  {'4','5','6','B'},

  {'7','8','9','C'},

  {'*','0','#','D'}

};

byte rowPins[ROWS] =

{9,8,7,6}; byte colPins[COLS]

= {5,4,3,2};

Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);

// - - - - - .

Variables --------

int totalStudents

= 0; int
```

```
presentCount =
0; int entered =
0;
String roll = "";
String studentInput
= ""; bool started =
false; bool
numberEntered =
false; void setup() {
lcd.init();
lcd.backlight();
lcd.clear();
  lcd.print("Press * to Start");
} void loop() {  char
key =
keypad.getKey();  if
(key) {
 // ---------- STEP 0: WAIT FOR START ----------
    if (!started) {       if (key
== '*') {         started = true;
lcd.clear();
lcd.print("Enter
No.Students");
lcd.setCursor(0,1);
      }
    }
   // ---------- STEP 1: ENTER TOTAL STUDENTS -----
-------- else if (!numberEntered) {
```

```
    if (key >= '0' && key <= '9')
{       studentInput += key;
lcd.print(key);
    }
 else if (key == '#') {  // Confirm
totalStudents = studentInput.toInt();
numberEntered = true;
        lcd.clear();
lcd.print("Total: ");
lcd.print(totalStude
nts); delay(1500);

        lcd.clear();
lcd.print("Enter Roll
No:");
    }    else if (key == '*') {
// Clear input
studentInput = "";
lcd.setCursor(0,1);
lcd.print("                ");
lcd.setCursor(0,1);
    }
   }
   // ---------- STEP 2: ATTENDANCE ----------
   else {
    if (key >= '0' && key
<= '9') {       roll += key;
lcd.setCursor(0,1);
lcd.print(roll);
    }
```

```
    else if (key == 'B') {
// Present
presentCount++;
entered++;
lcd.clear();
lcd.print("Roll ");
lcd.print(roll);
lcd.setCursor(0,1);
lcd.print("Present");

      delay(1200);
nextStudent();
    }
    else if (key == 'A') {
// Absent        entered++;
lcd.clear();
lcd.print("Roll ");
lcd.print(roll);
lcd.setCursor(0,1);
lcd.print("Absent");
delay(1200);
nextStudent();
    }
    else if (key == 'D') {  // Show Result     lcd.clear();
float percentage = ((float)presentCount / totalStudents) *
100;      lcd.print("Attendance:");
lcd.setCursor(0,1);        lcd.print(percentage);
lcd.print("%");
    }
```

```
    else if (key == '*') {  // Clear roll
input        roll = "";
        lcd.setCursor(0,1);
        lcd.print("                ");
        lcd.setCursor(0,1);
      }
    }
  }
}
void nextStudent() {
  roll = "";
lcd.clear();   if
(entered <
totalStudents) {
lcd.print("Enter Roll
No:");
  } else {
lcd.print("Press D
Result");
  }
}
```

CHAPTER 6

RESULT

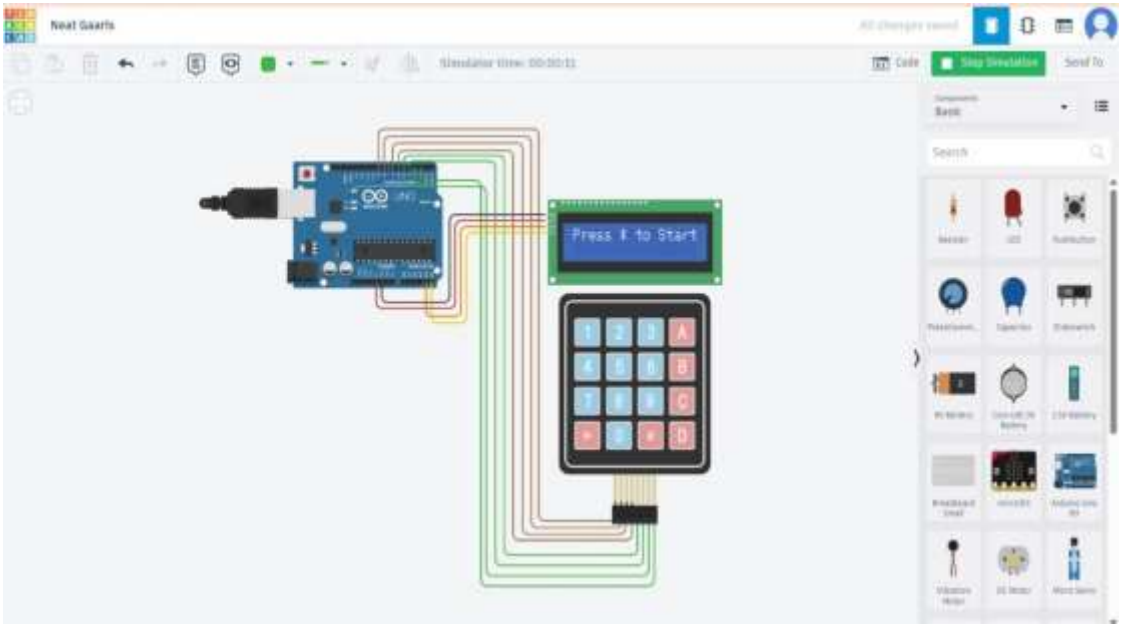

Fig 6.1

Fig 6.2

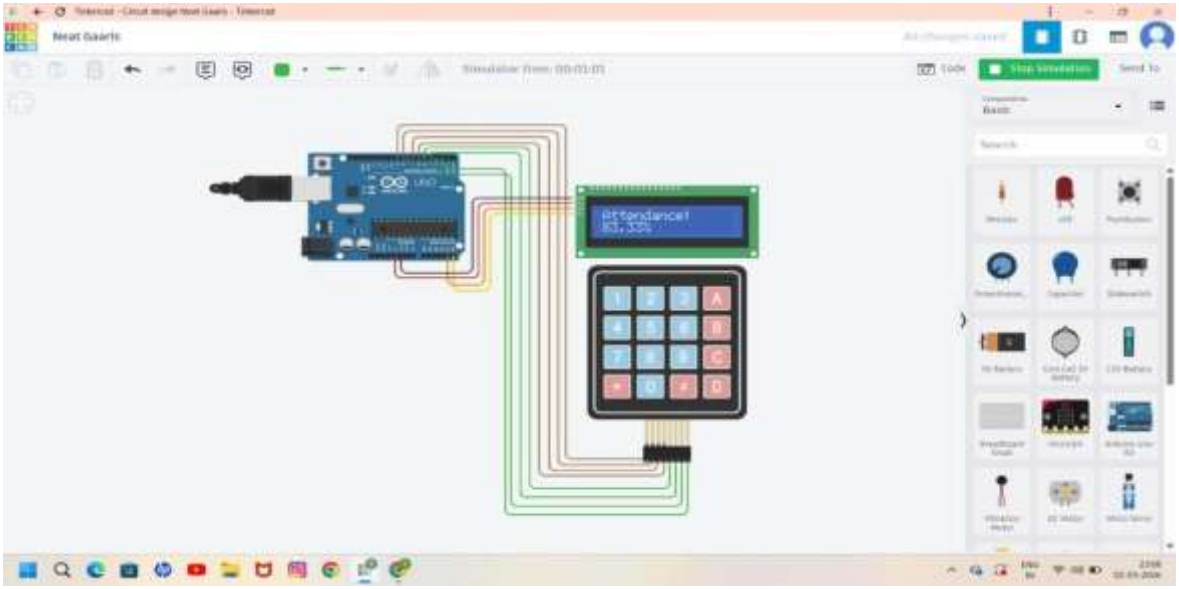

Fig 6.3

CHAPTER 7

SDG IMPACT ANALYSIS

The Digital Attendance System directly supports Sustainable Development Goal 4 (SDG 4): Quality Education, which aims to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. Efficient attendance monitoring is an essential component of maintaining academic discipline and improving student engagement. By automating the attendance process, the system enhances classroom management and reduces administrative burden on teachers.

The project contributes to SDG 4 in several ways. First, it improves instructional efficiency by saving valuable teaching time that would otherwise be spent on manual attendance recording. Second, it minimizes human errors in marking and calculating attendance percentages, ensuring accurate academic records. Third, the system promotes digital transformation in educational environments by integrating embedded systems technology into routine administrative tasks.

Additionally, the Digital Attendance System encourages the adoption of low-cost technological solutions in schools and colleges, especially in resource-constrained institutions. Since the system is built using affordable and widely available components like Arduino, LCD, and keypad, it can be implemented in both urban and rural educational settings.

Overall, this project strengthens educational monitoring systems, supports data accuracy, and enhances institutional efficiency, thereby contributing to improved learning outcomes and the broader objective of achieving SDG 4 – Quality Education.

CHAPTER 8

SUMMARY

The Digital Attendance System is an Arduino-based automated solution designed to simplify classroom attendance management. It allows the user to enter the total number of students, input individual roll numbers, and mark each student as present or absent using a keypad, while displaying all information on a 16×2 LCD screen. The system automatically counts attendance and calculates the final attendance percentage once all entries are completed. By reducing manual effort, minimizing errors, and providing instant results, the project offers a simple, cost-effective approach to improving academic efficiency in support of SDG 4 – Quality Education.

CHAPTER 9

DISCUSSION

Fig 9.1

pmk-fstd-isu
sanlya
Vinay P
Sinchana
ESHWAR
Naveen
DEEPA SREE
Arquam
Anusha
Manoj BK
Divya
Naveen
3 others
You
Return to top
Amar
pmk-fstd-isu
sanlya
Anusha
DEEPA SREE
Manoj BK
ESHWAR
Kumuda N gowda
Divya
Vinay P
Arquam
You
6 others
Kumuda N gowda
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Fig 9.2